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Personal Loan Approval/Deny Prediction

A Comparative Study of Machine Learning Classification Models

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01 INTRODUCTION

Personal loan approval has become an important application of machine learning because lending decisions must balance speed, consistency, and risk control. In practice, financial institutions process large volumes of applications and must distinguish between qualified and unqualified borrowers while minimizing costly misclassification errors. This paper examines whether machine learning models can improve the prediction of personal loan approval outcomes relative to more traditional linear approaches. Using a Kaggle dataset of 5,000 observations, the study analyzes borrower demographic, financial, and banking-related characteristics to classify whether a loan is approved or denied. A key challenge in the dataset is its strong class imbalance, with roughly 90% of observations belonging to the denied class, making overall accuracy an unreliable measure of model quality. The analysis also incorporates feature engineering, including city-level income variables and an income-to-median-income ratio, to better capture each borrower's financial context. The paper first presents the data and feature engineering process, then reviews the relevant literature, explains the models tested, compares their results, and concludes with the best-performing approach and its practical implications.

02 DATA

2.1 The Original Dataset

A dataset containing 5000 observations was acquired from Kaggle. It had the following 13 features: ID, Age, Experience, Income, Family (# of members), CCAvg (average credit score), Education, Mortgage (1 = taken, 0 = not taken), Securities.Account (1 = has securities account), CD.Account (1 = has CD.Account), Online (1 = has online banking), Credit Card (1 = has credit card), Zip Code. The target variable, "Personal.Loan", is binary where 1 = loan is approved, and 0 = loan is denied.

2.2 Feature Engineering

In order to get better results, a couple of changes were made to the original data. First, there was the issue that Zip codes are stored as integers, but they represent categorical (nominal) data, not numerical (continuous/float) values. In order to solve that, external data was used to turn zip codes into city names using the XLOOKUP function. Second, using the city names, median income per city was added to the data, to add information on the economic state of the location where the request was made. Third, "Income" was replaced by "Income / Median in City". This ratio adds context to the observation's financial position. For example, $1 in New York can be very different to $1 in Mississippi. Finally, "ID" was taken out as it does not add any useful information. The updated dataset had the following features: Age, Experience, Family, CCAvg, Education, Mortgage, Securities.Account, CD.Account, Online, CreditCard, City, Median income per city, Income / Median in City.

2.3 Observations

The dataset had very high class imbalance. With 90% of the loans being rejected, the accuracy of supervised ML models can be misleading. A model that predicts "rejected" every time would still get an accuracy of 90%. This is why to evaluate performance of models, ROC AUC and F1 score will be given more importance. Additionally, the data was unstandardized, which can lead to certain features to dominate due to scale differences, leading to less accurate model performance. For that reason, the data was standardized on Excel before being imported onto python.

03 PREVIOUS FINDINGS

Loan approval prediction sits at the intersection of finance and machine learning because it is not only a technical classification task, but also a credit-allocation decision made under uncertainty. In this setting, the literature shows that the central question is not simply which model produces the highest overall accuracy, but which approach best balances predictive performance, interpretability, and reliable identification of the minority class. Prior studies collectively suggest that credit outcomes are shaped by both linear relationships and more complex interactions, which is why the present project compares interpretable linear baselines, regularized linear models, tree-based and kernel methods, and neural networks in a single personal-loan approval setting.

A first stream of research establishes why simple linear models remain an important starting point. Li, Mickel, and Taylor (2018) study how loan approval can be framed as a disciplined binary classification problem rather than an ad hoc lending judgment. Using a large U.S. Small Business Administration dataset, they emphasize a logistic-regression-style baseline and show that structured borrower data can be translated into a reproducible approval rule. Their main contribution is not that one basic linear model is always best, but that credit approval should begin with a transparent benchmark whose coefficients can be interpreted economically. This matters directly for the present project because it justifies including OLS-style and logistic-style baseline models as the first point of comparison. In lending contexts, interpretability is not a minor convenience; it is part of the decision value of the model itself.

At the same time, the literature also shows why linear baselines may be too restrictive when borrower risk depends on correlated predictors, threshold effects, or variable interactions. This is where regularized linear methods become important. Li et al. implicitly motivate that next step by showing the usefulness of a disciplined linear starting point, while the broader logic of financial tabular data suggests that Ridge, Lasso, and Elastic Net can improve stability when predictors overlap or contain noise. In other words, regularization is not included merely for technical variety. It allows this project to test whether a simpler and more interpretable decision rule can remain competitive once shrinkage and feature selection are applied. That makes regularized models a meaningful middle ground between plain linear baselines and more flexible machine-learning approaches.

A second stream of research argues more strongly for non-linear methods. Brown and Mues (2012) ask which classifiers perform best when credit-scoring data are imbalanced, using five real-world datasets to compare logistic regression, neural networks, decision trees, random forests, gradient boosting, and support vector machines. Their main finding is that ensemble methods, especially random forests and gradient boosting, perform particularly well and remain more robust as imbalance worsens. Zhou, Shen, and Ballester (2023) reach a similar conclusion in a different setting by studying credit scoring for Chinese small firms with imbalanced data that include financial, non-financial, and macroeconomic variables. Their two-stage approach combines SMOTE and random forest feature construction, then uses dominance analysis to identify influential predictors; they find that this framework outperforms competing models and that broader contextual variables materially improve prediction. Taken together, these studies suggest that single linear boundaries may miss important structure in credit data. That is why this project includes tree-based models and SVMs: when approval outcomes depend on interactions, non-linear risk patterns, or flexible margins between accepted and rejected borrowers, these models may capture information that linear baselines cannot.

More recent comparative evidence reinforces this point, while also warning against simplistic claims that one model always dominates. Alagic et al. (2024) examine whether multiple machine-learning classifiers can improve loan approval and credit-risk prediction in a smaller, survey-based dataset. They compare several algorithms and find that ensemble methods such as XGBoost and random forest again perform strongly. However, their study also highlights an important limitation in the literature: performance rankings are sensitive to dataset size, variable design, and evaluation strategy. Li et al. rely on a large administrative lending dataset, Brown and Mues compare multiple real-world credit datasets, Zhou et al. focus on Chinese small firms, and Alagic et al. work with a more specialized survey-based sample. As a result, the literature does not support the claim that there is one universally best classifier for every lending problem. Instead, it points to a more useful conclusion: flexible non-linear methods often outperform simpler models when the data contain richer structure, but that advantage must be demonstrated within the specific dataset and evaluation framework being used.

That evaluation framework is especially important here because the present dataset is roughly 90/10 imbalanced. In such a setting, accuracy can be misleading: a model can appear strong by mostly predicting the majority class while failing to identify the class of greatest analytical interest. Saito and Rehmsmeier (2015) address exactly this issue by asking how binary classifiers should be evaluated under class imbalance. They show that ROC-based summaries alone can overstate model quality and that precision-recall analysis is often more informative when minority-class detection matters. Their study matters directly for this project because it clarifies why model comparison cannot rely on accuracy by itself. For a 90/10 approval dataset, recall is necessary to assess whether the model is actually identifying minority outcomes, F1-score helps evaluate the balance between missed cases and false alarms, and ROC-AUC remains useful as a broader ranking metric. Together, these measures provide a more credible assessment of lending-model performance than overall accuracy alone.

Viewed as a whole, the literature reveals a clear research gap. Many prior studies focus on default rather than approval, evaluate only a narrow subset of models, or rely on datasets that are either highly specialized or not directly comparable to a personal-loan approval context. Just as importantly, the literature often separates the interpretability question from the performance question, instead of testing both within one coherent framework. Previous work also does not always treat class imbalance as a central design issue, even though imbalanced outcomes are common in real credit decisions. This project responds to those gaps by comparing interpretable linear baselines, regularized models, tree-based and kernel methods, and neural networks within one dataset and under an evaluation framework that explicitly prioritizes minority-class performance. Its contribution is therefore not simply to apply familiar algorithms, but to examine whether the gains from added complexity are large enough to justify moving beyond transparent financial decision rules.

Based on the literature, the most plausible expectation is that flexible non-linear models—particularly ensemble tree methods, and possibly SVMs—will outperform simple linear baselines in predictive performance because they are better able to capture interactions and non-linear approval boundaries. At the same time, linear and regularized linear models are expected to remain valuable benchmarks because they offer greater interpretability and may still perform reasonably well if the underlying structure of the data is not overly complex. Neural networks may provide additional flexibility, but the literature does not justify assuming that they will dominate on structured tabular credit data. The overall hypothesis, therefore, is that ensemble or other non-linear methods will achieve the strongest recall- and F1-oriented performance, while regularized linear models may offer the most attractive tradeoff between transparency and predictive quality.

04 **METHODOLOGY**

4.1 Unsupervised Machine Learning

To better understand the data, the following Unsupervised Machine Learning:

4.1.1 K-Means Clustering

To determine the optimal number of clusters (K) to use, two methods were used. First, elbow screen plot. As seen in Appendix 1, the highest drop in WCSS is seen when increasing K from 1 to 2. The second method was calculating the silhouette score, which peaked at 0.637 at K=2. Both methods indicate that the optimal number of clusters is 2. Appendix 2 demographic data, with “Age”, “Experience”, and “City”, being the axes. Appendix 3 visualizes financial data, with "Mortgage", "Income / Median in City", and "CCAvg" as axes. The demographic data shows clearer group separation, suggesting more meaningful and distinguishable clusters. Meanwhile, the financial data, which can be more complex, is unclear, with significant overlap between groups indicating weak separation. This can be due to high variability in financial data, overlap between observations, financial data having less distinct patterns compared to demographic data.

4.1.2 Fuzzy Clustering

Two fuzzy clustering models were ran, the first with fuzzy coefficient m = 1.5 (Appendix 4), and the second with m = 3 (Appendix 5). The fuzzy coefficient controls how strongly each observation belongs to a cluster. When m = 1.5, the clustering is less fuzzy, so points are assigned more strongly to one group, creating a sharper and more distinct boundary. When m = 3, memberships become softer, so more observations share partial membership across clusters, making the transition between groups appear more gradual and blurred. This reveals that the data does contain an underlying cluster structure, since the overall pattern remains similar in both plots. However, it also shows that the clusters are not perfectly distinct, as many observations fall into overlapping or ambiguous regions, especially near the center.

4.1.3 Hierarchical Clustering

Hierarchical clustering produced a different result from K-means clustering, identifying an optimal k = 3 rather than k = 2. This suggests that the data may contain a more layered structure that hierarchical methods can capture better than K-means, which assumes more compact and spherical clusters. In Appendix 6, the dendrogram shows how observations merge step by step, and the larger vertical distances between the final branches indicate that cutting the tree at three groups is reasonable. In Appendix 7, the PCA visualization confirms this result by showing three relatively distinct clusters, mainly separated along the first principal component. Although some overlap remains, the hierarchical clustering results suggest a more detailed grouping structure than the simpler two-cluster solution found with K-means.

4.2 Linear Probability Model (OLS Baseline)

Ordinary Least Squares (OLS) is a fundamental regression model used to estimate the linear relationship between a dependent variable and independent variables by minimizing the sum of squared residuals. Due to its simplicity and transparency, OLS is used as a baseline linear probability model to predict loan approval outcomes. The model was trained on all features without regularization, and continuous outputs were converted into binary classifications using a threshold of 0.5.

On the test set, the model achieved strong accuracy and a high ROC AUC. However, recall was low relative to precision, indicating that the model fails to identify a large proportion of eligible borrowers. This reflects the limitations of applying a linear model to an imbalanced classification problem.

Importantly, residual analysis (see Appendix A.9) reveals clear patterns rather than random dispersion, indicating violations of key OLS assumptions such as homoscedasticity. In addition, the distribution of predicted values shows outputs outside the logical [0,1] probability range (see Appendix A.8), highlighting a fundamental limitation of using OLS for classification tasks.

These results suggest that the relationship between predictors and loan approval is not purely linear. As a result, more flexible models, such as SVM, are better suited to capture the complex and non-linear patterns present in the data. Overall, OLS provides a useful benchmark due to its transparency and ease of interpretation, but its performance highlights the need for more flexible models to better capture underlying patterns in loan approval decisions.

4.3 Logistic Regression

The logistic regression model is implemented using a logit link function to model a binary outcome, where the dependent variable indicates whether a customer accepts a personal loan. This approach is more appropriate than ordinary least squares because it ensures predicted values remain between zero and one and directly models probabilities for classification. The model estimates the relationship between input features and the likelihood of loan acceptance through a linear combination of predictors transformed into probabilities.

The dataset is split into 80 percent for training and 20 percent for testing, with stratification to preserve the class distribution. The model is trained on the training set and evaluated using accuracy, precision, recall, F1 score, and ROC AUC, which provide a more complete assessment given the class imbalance.

The results show an accuracy of 0.947 and a precision of 0.831 in Appendix 24, indicating that predicted positives are generally reliable. However, recall is 0.562, meaning that a significant number of actual loan acceptors are not identified. This is confirmed by 42 false negatives in the confusion matrix. The F1 score of 0.671 reflects this imbalance, while the ROC AUC of 0.950 indicates good ranking ability. Overall, the model performs well on the majority class but struggles to capture the minority class.

4.4 Regularized Regression: Ridge, Lasso and Elastic Net

Regularized regression models are designed to improve the performance of linear models in the presence of multicollinearity and noisy predictors, which are common in financial datasets. By adding a penalty term to the loss function, these models reduce overfitting and improve generalization to unseen data.

In this section, three regularized regression techniques are implemented: Ridge, Lasso, and Elastic Net. While all three models are based on the same linear framework, they differ in how they penalize coefficients, which affects both model complexity and interpretability. Given the imbalanced nature of the dataset, model performance is primarily evaluated using ROC AUC and F1-score, as these metrics better capture classification performance than accuracy alone.

4.4.1 Ridge

Ridge regression introduces an L2 penalty to the regression objective function, shrinking coefficients toward zero without eliminating any variables. This makes it particularly effective when predictors are highly correlated, as it stabilizes coefficient estimates while retaining all available information.

The model was trained using 5-fold cross-validation, resulting in an optimal regularization parameter of $\alpha = 0.001$. On the test set, Ridge regression achieved an MSE of 0.164652 and an R^2 of 0.420683. In terms of classification performance, the model obtained a ROC AUC of 0.7147 and an F1-score of 0.4706, with a precision of 0.6923 and a recall of 0.3529.

These results indicate that Ridge provides stable and consistent predictions, with relatively strong discrimination ability as reflected in the ROC AUC. However, the moderate F1-score highlights an imbalance between precision and recall, with the model favoring precision over recall. This suggests that while Ridge is effective at avoiding false positives, it fails to identify a significant proportion of true positive cases. Since all 54 features are retained, the model does not improve interpretability, but it ensures robustness in the presence of correlated variables.

4.4.2 Lasso

Lasso regression applies an L1 penalty to the loss function, which allows it to shrink some coefficients exactly to zero and therefore perform automatic feature selection. This leads to a model that focuses only on the most relevant predictors.

Using cross-validation, the optimal regularization parameter was found to be $\alpha = 0.1$. On the test set, Lasso achieved an MSE of 0.164780 and an R^2 of 0.420239. The model obtained a ROC AUC of 0.7141 and an F1-score of 0.4688, with precision of 0.6909 and recall of 0.3510. The number of features was reduced from 54 to 39.

Despite removing a substantial number of variables (see Appendix A.10), Lasso maintains performance very close to Ridge, indicating that the eliminated features contribute little additional predictive power. The consistency of key variables such as Income, CD.Account, Education, and CCAvg across all three models suggests that these features are strong and stable predictors of loan approval. The ROC AUC remains almost unchanged, confirming that the model's ability to distinguish between classes is preserved. Similarly, the F1-score remains comparable, suggesting that feature reduction does not significantly affect the balance between precision and recall.

These results highlight Lasso's effectiveness in simplifying the model without sacrificing performance. However, like Ridge, it continues to show relatively low recall, indicating difficulty in identifying positive loan approval cases.

4.4.3 Elastic Net

Elastic Net combines both L1 and L2 penalties, allowing it to benefit from both feature selection and coefficient shrinkage. The model introduces an additional hyperparameter, the l1_ratio, which determines the contribution of the L1 (Lasso) and L2 (Ridge) penalties. This makes Elastic Net particularly useful when predictors are correlated, as it avoids the limitations of Lasso in selecting only one variable from a group.

The model was tuned using cross-validation, yielding optimal parameters of $\alpha = 0.1$ and l1_ratio = 0.5, indicating a balanced contribution from both penalties. On the test set, Elastic Net achieved an MSE of 0.164651 and an R^2 of 0.420690. In classification terms, it obtained a ROC AUC of 0.7151 and an F1-score of 0.4716, with precision of 0.6935 and recall of 0.3539. The model retained 42 out of 54 features.

Elastic Net provides the strongest overall performance among the three models, although the improvement is marginal. The slightly higher ROC AUC suggests a marginally better ability to distinguish between classes, while the improved F1-score reflects a slightly better balance between precision and recall. The number of retained features lies between Ridge and Lasso, confirming its hybrid nature.

The l1_ratio of 0.5 indicates that both feature selection and coefficient shrinkage are equally important for this dataset. This suggests that the data benefits from reducing redundancy while still preserving information from correlated variables. As with the other models, recall remains relatively low, indicating that the main limitation lies in the dataset rather than the modeling approach. The selected hyperparameters and corresponding training performance (see Appendix A.11) further confirm the stability of the model across different regularization settings.

4.5 Support Vector Machine (SVM)

Support Vector Machines (SVM) are supervised learning models used for classification that construct an optimal decision boundary by maximizing the margin between classes. This margin represents the distance between the separating hyperplane and the closest observations, and maximizing it improves generalization performance on unseen data. Unlike linear models, SVM can capture non-linear relationships through kernel functions, making it well suited for financial datasets where interactions between variables are complex.

In this project, an SVM model with a Radial Basis Function (RBF) kernel was implemented to allow for non-linear decision boundaries. Hyperparameter tuning was performed using 5-fold cross-validation over 16 parameter combinations. The optimal configuration was C=10, γ=0.1, with a cross-validation accuracy of approximately 97.10%, reflecting a balance between model flexibility and generalization. The hyperparameter search results (see Appendix A.13) show that the highest ROC-AUC is achieved at C=10, γ=0.1 and C=100, γ=0.01confirming the robustness of the selected configuration.

On the test set, the SVM model achieved an accuracy of 97.60%, with a precision of 0.9091, recall of 0.8333, and an F1-score of 0.8696, indicating a strong balance between correctly identifying loan approvals and limiting classification errors. The ROC AUC of 0.9885 further confirms the model's excellent discriminative ability. The precision–recall trade-off analysis (see Appendix A.12) shows that performance remains stable across thresholds, with an equilibrium point around a threshold of approximately -0.30 where precision and recall are balanced. This threshold shows the model is conservative by default and tend to predict class 0.

The confusion matrix shows that the model correctly classified 896 true negatives and 80 true positives, while producing only 8 false positives and 16 false negatives. The low number of false positives is particularly important in a financial context, as it minimizes the risk of approving loans to unqualified applicants, while the relatively high recall indicates that a large proportion of eligible borrowers are correctly identified. The threshold analysis (see Appendix A.12) further highlights that increasing the threshold improves precision at the expense of recall, while lowering it increases recall but introduces more false positives, reinforcing the trade-off inherent in classification decisions.

These results suggest that the relationship between the predictors and loan approval is not purely linear. The strong performance of the SVM, particularly with a non-linear RBF kernel, indicates that more complex patterns and interactions between variables play an important role in determining outcomes. However, SVM remains less interpretable than linear models, as it does not provide direct information on the contribution of individual predictors.

4.6 KNN – K Nearest Neighbours

K-Nearest Neighbors (KNN) classifies each observation by identifying the K most similar training cases using Euclidean distance and assigning the majority class among those neighbors. Because the dataset is already standardized, all features contribute proportionally to the distance calculation without scaling. The optimal K was selected by looping over values from 1 to 20 and choosing the value minimizing the misclassification rate on the test set, which identified K = 3 (Appendix A.15). To truly understand the precision and recall tradeoff for different Ks, three different K values were tested (K= 3, 5, 7).

The observation was the following: as K increases, precision improves while recall falls (Appendix A.16). At K = 3, recall is 0.573 with precision sitting at 0.917. When observing the precision/recall tradeoff at a higher K of 7, recall drops to 0.469 while precision reaches 0.957. This inverse relationship is a reflection of the class imbalance in the data because at larger K values, the denied majority outvotes the approval minority, pushing predictions toward denial regardless of the applicant's actual characteristics. No K value resolves both problems simultaneously. K = 3 was retained as the best available option because recall is the priority in a loan approval context, where failing to identify a creditworthy applicant carries a higher cost than a false approval. The selected model achieved accuracy of 0.954, F1 of 0.705, and AUC of 0.887 (Appendix A.17).

Overall, the KNN model performance confirm that distance-based classification is not very effective on partitioning on a heavily imbalanced dataset.

4.7 CART – Classification and Regression Tree

This study employs a Classification and Regression Tree (CART) to predict personal loan approval. Unlike logistic regression, which assumes a linear relationship between predictors, CART partitions the feature space using binary splits, minimizing entropy at each node. This makes it well-suited to the dataset used, which contains continuous variables such as income and credit card spending alongside binary indicators like CD account ownership and securities account status. CART is one of the better suited models due to the *relationship* between these predictors and loan approval, and *its* unlikeliness to follow a simple linear pattern. The model has a train/ test split of 80/20 and uses stratified sampling to mirror the 90/10 distribution into the test/train subsets.

Throughout the configuration of CART to our dataset, we tested 4 configurations to assess the effect of tree depth and pruning on model performance. Our first test and baseline model for other CART configurations (depth=5, no pruning) produced a test AUC of 0.982 and recall of 0.865. Upon observation of terminal nodes, some contained as few as two observations. A terminal node based off only two data points was deemed too small to be reliable. If those two observations happen to be noise or outliers, the rule they produce will not hold on to new data going forward in practice. Hence, it was classified as a fragile split.

To ensure that the model doesn't overfit to patterns that may not generalize beyond the scope of the training set, a minimum observations per leaf configuration was applied in the form of min_samples_leaf = 10. This forces each leaf to contain at least 10 observations which in turn forces the model to make rules based on more meaningful patterns rather than possible noise, slightly affecting performance at a depth of 5 in the name of more reliable generalization.

Depth was then varied across three and seven to examine the tradeoff between interpretability and performance (Appendix A.18). Depth = 3 reduced recall to 0.521, demonstrating that excessive pruning is as costly as none at all. Reducing depth to three limits the tree to only three levels of splits, forcing it to classify most applicants using broad rules. With only eight possible terminal nodes to partition 5,000 observations, the model cannot capture the finer combinations that distinguish approved applicants from denied ones.

The selected configuration, depth = 7 with min_samples_leaf = 10, delivered the strongest performance: accuracy of 0.977, recall of 0.906, F1 of 0.883, and AUC of 0.978. The train-test AUC gap was 0.019, below the 0.02 threshold for meaningful overfitting, and 10-fold cross-validation yielded a mean accuracy of 0.970 with a standard deviation of 0.008, confirming stable generalization across subsets of the data. At depth seven, the tree has sufficient levels to capture the interaction between variables that characterizes approved applicants, while min_samples_leaf = 10 ensures each decision rule is grounded in enough observations to be reliable. The additional depth beyond five allows the model to resolve borderline cases that shallower configurations classified incorrectly, particularly among applicants whose income relative to their city median falls near the approval threshold.

Following model optimization, we observed feature importance scores from the selected model that reveal that our two engineered geographic variables of Median Income per City and Income/Median in city account for around 58% of total predictive weight (Appendix A.19). This is more than all the original dataset variables combined. Credit card spending (CCAvg) contributed 22.8% and education level 11.7%.

Four variables including age, mortgage, securities account, and online banking received zero importance, meaning the tree never used them to make a classification decision.

4.8 Random Forest & Gradient Boosting

The Random Forest model shows a clear improvement in performance across all key metrics. With 50 trees, the model achieves an accuracy of 0.982, Appendix 27 , which is substantially higher than logistic regression. Precision increases to 0.943, indicating that predicted positive cases are highly reliable. More importantly, recall rises to 0.865, meaning that the model correctly identifies a much larger proportion of actual loan acceptors. The confusion matrix reflects this improvement, with 83 true positives and only 13 false negatives. This represents a major reduction in missed opportunities compared to logistic regression. The F1 score of 0.902 demonstrates a strong balance between precision and recall, while the ROC AUC of 0.993 indicates excellent discrimination between classes. These results confirm that the model is highly effective at capturing patterns in the data.

When increasing the number of trees from 50 to 300, the performance remains very stable. Accuracy slightly decreases to 0.981, while precision remains unchanged at 0.943. Recall decreases marginally to 0.854, with one additional false negative compared to the 50-tree model. The F1 score decreases slightly to 0.896, while the ROC AUC improves from 0.993 to 0.995. This suggests that adding more trees improves the quality of probability estimates rather than significantly enhancing classification performance. The confusion matrices for both configurations are nearly identical, confirming that the model has already captured most of the underlying structure of the data with a relatively moderate number of trees.

The behavior of the Random Forest model with only 2 trees highlights the importance of ensemble size. In this configuration, accuracy drops to 0.953, which is still relatively high due to the dominance of the majority class. However, recall falls sharply to approximately 0.55, indicating that the model fails to detect a large number of loan acceptors. The confusion matrix shows 43 false negatives, which is comparable to the performance of logistic regression. Precision remains high at approximately 0.93, meaning that when the model predicts a positive case, it is usually correct. However, the F1 score decreases to approximately 0.69, reflecting the poor balance between precision and recall. These results demonstrate that a small number of trees is insufficient to capture the complexity of the data and that the strength of Random Forest lies in aggregating a large number of diverse decision trees.

The Gradient Boosting model achieves the best overall performance among all models. It reaches an accuracy of 0.986, Appendix 24, with both precision and recall equal to 0.927. This balance between precision and recall is particularly significant, as it indicates that the model is equally effective at identifying loan acceptors and avoiding false positives. The confusion matrix shows 897 true negatives, 89 true positives, and only 7 false positives and 7 false negatives. This represents a substantial improvement compared to both logistic regression and Random Forest. The F1 score of 0.927 confirms that the model maintains a strong balance between the two types of errors, while the ROC AUC of 0.994 indicates near-perfect discrimination between classes. Compared to the Random Forest model with 50 trees, Gradient Boosting reduces the number of false negatives from 13 to 7, while maintaining a low number of false positives. This reduction in missed loan acceptors is particularly important in a business context, where failing to identify potential customers can lead to lost opportunities. The results demonstrate that the sequential nature of boosting, where each new model focuses on correcting previous errors, leads to improved performance on difficult cases.

4.9 Neural Network

Given the 90/10 class imbalance, accuracy alone is an unreliable performance signal. All supervised models are evaluated on five metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Recall and ROC-

AUC are treated as primary criteria, as the cost of a false negative — approving a loan that defaults — is operationally more severe than a false positive.

The Convolutional Neural Network model presents a different set of results. In its standard configuration, the model achieves an accuracy of 0.946, which is comparable to logistic regression. The precision is 0.776, indicating that predicted positive cases are correct approximately 78 percent of the time, Appendix 22. The recall is 0.615, which means that the model identifies around 61 percent of actual loan acceptors. The confusion matrix shows 887 true negatives, 59 true positives, 17 false positives, and 37 false negatives. The F1 score of 0.686 reflects a moderate balance between precision and recall, while the ROC AUC of 0.925 indicates that the model has a good ability to distinguish between classes, although it is lower than that of the tree-based models. These results suggest that while the CNN is capable of capturing some non-linear relationships, it does not perform as well as Random Forest or Gradient Boosting on this structured dataset.

When the model is trained for a longer duration, performance improves. Accuracy increases slightly to 0.953, while recall rises significantly to 0.719. Precision remains stable at approximately 0.775, indicating that the model maintains a similar level of reliability in its positive predictions. The F1 score increases to 0.746, reflecting a better balance between precision and recall. The number of true positives increases from 59 to 69, while false positives increase slightly from 17 to 20. The ROC AUC also improves from 0.925 to 0.948, suggesting better overall discrimination. These changes indicate that additional training allows the model to capture more patterns in the data, particularly those associated with the minority class.

However, when modifying the architecture by using sigmoid activation functions in hidden layers, the performance deteriorates significantly. At a classification threshold of 0.5, accuracy drops to 0.844, while precision decreases to 0.354. Recall remains relatively high at 0.760, but this comes at the cost of a large number of false positives. The confusion matrix shows 133 false positives, indicating that many non-loan customers are incorrectly classified as loan acceptors. The F1 score decreases to 0.483, reflecting a poor balance between precision and recall. Across different thresholds, the model exhibits strong sensitivity, with lower thresholds increasing recall but drastically reducing precision, and higher thresholds improving precision at the expense of recall. This instability suggests that the model is highly dependent on hyperparameter choices and is less robust than the tree-based approaches.

05 RESULTS

Throughout this project 10 models were tested across four overarching methodologies: linear baselines, regularized regression, non-linear classifiers and ensemble methods. The primary metrics used to evaluate models were recall, F1-score, and ROC-AUC. The results table shows a clear performance hierarchy (Appendix A.0)

The linear models (OLS, Logistic Regression, Ridge, Lasso, and Elastic Net) all have poor recall despite having an accuracy of 90%+. As established in Section 2, this reflects the 90/10 class imbalance: a linear decision boundary cannot cleanly separate the approval minority from the denial majority when the classes overlap in feature space.

The performance of the non-linear single models showed significant improvement. The Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel, using parameters C=10 and γ=0.1, achieved a recall of 0.833, an F1 score of 0.870, and an AUC of 0.989. This substantial improvement indicates the kernel's ability to create a flexible boundary between accepted and rejected applicants.

The Classification and Regression Tree (CART), with a depth of 7 and a minimum leaf sample size of 10, performed similarly, with a recall of 0.906, an F1 score of 0.883, and an AUC of 0.978. As discussed in Section 4.7, the feature importance analysis showed that the two newly created geographic variables together accounted for 58% of the total predictive weight, more than all the original dataset variables combined, thus directly confirming the feature engineering process described in Section 2.

KNN (K=3) lags behind both at recall of 0.573 and AUC of 0.887. As discussed in Section 4.6, the class imbalance structurally limits distance-based classification because the denied majority dominates neighborhood votes at every tested K value. The convolutional neural network (CNN) demonstrated a recall of 0.719 and an area under the curve (AUC) of 0.948. This performance surpassed that of the linear models, yet it was inferior to the support vector machine (SVM) and classification and regression tree (CART) models. This is consistent with convolutional architectures being better suited to image or time-series data than rows of borrower characteristics.

Gradient Boosting deliver the strongest results out of the 10 models. It performance metrics of an accuracy of 98.6%, recall of 0.927, F1 of 0.927, and AUC of 0.994 makes it the obvious recommendable model for the dataset. The advantage of Gradient Boosting over a single tree like CART comes from each successive tree being trained to correct the errors of the previous one, with greater weight placed on previously misclassified observations. In a 90/10 imbalanced dataset, this means the model iteratively focuses attention on the approval minority rather than defaulting toward the denial majority, which explains why recall improves beyond what any single model achieves.

The sequential ensemble structure also provides implicit regularization, so the performance gains do not come at the cost of overfitting. Therefore, Gradient Boosting is selected as the best-performing model. Its recall and F1 lead every other model by at least three percentage points, and its AUC of 0.994 is the highest observed across all configurations tested. The interpretability tradeoff relative to CART is noted but, in a setting, where the priority is reliable identification of creditworthy applicants, the performance advantage on recall and F1 is the more relevant consideration than CART's interpretability.

06 **CONCLUSION**

This study set out to determine whether machine learning models can improve personal loan approval prediction relative to traditional linear approaches, and the results provide a clear answer. Across ten models tested on a 5,000-observation dataset with a 90/10 class imbalance, the gap between linear and non-linear methods was substantial and consistent. OLS, Ridge, Lasso, and Elastic Net all failed to identify more than 35% of actual approvals despite achieving accuracy above 92%, confirming that a linear decision boundary is structurally insufficient for this classification task. Gradient Boosting achieved recall of 92.7%, F1 of 92.7%, and AUC of 0.994, confirming the hypothesis from the literature review that ensemble methods would outperform linear baselines by a practically meaningful margin.

Three findings from this study are worth emphasizing. First, the feature engineering process contributed more predictive value than any variable in the original dataset. Income relative to city median was the single most important predictor in CART at 41.4% importance and the strongest coefficient in every regularized linear model. A dollar earned in a low-income city implies a different financial position than the same dollar in a high-cost urban area, and the unsupervised analysis reinforced this, finding clearer borrower segment separation along demographic dimensions than financial ones. Second, the class imbalance is not simply a technical inconvenience but the central design challenge of the problem. Models that ignored it consistently produced misleading accuracy figures while failing on recall, the share of creditworthy applicants correctly identified. Third, interpretability and performance are not always at odds. CART achieved recall of 90.6% and AUC of 0.978, and a retail lender that values explainability could deploy it with only a modest tradeoff relative to Gradient Boosting.

The study has some limitations. The dataset does not represent any specific institution's lending portfolio, meaning feature relationships may not transfer to real-world deployment without retraining. SMOTE and cost-sensitive learning were not applied and remain opportunities for future work.

The geographic enrichment relied on median income estimates that may not reflect within-city variation accurately. Future work could apply the best-performing models to a real institutional dataset, test ensemble stacking, and incorporate macroeconomic variables such as interest rates or unemployment at the city level. The results here suggest that thoughtful feature engineering combined with flexible non-linear models offers a practically useful and statistically credible approach to personal loan approval prediction.

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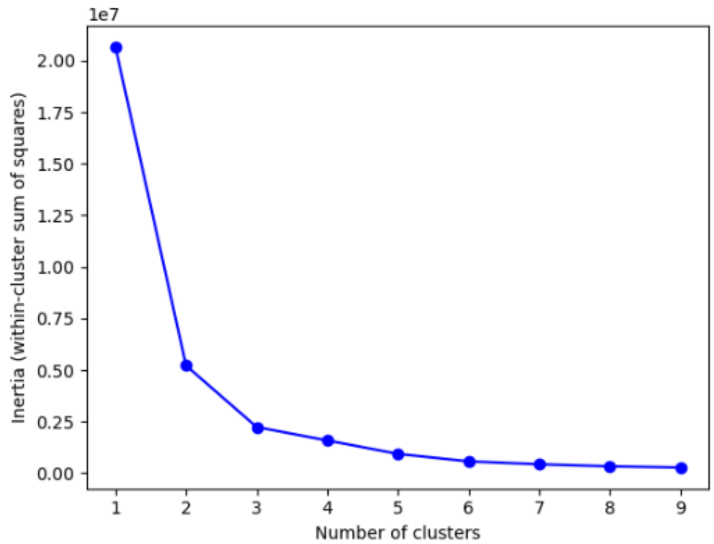
APPENDIX

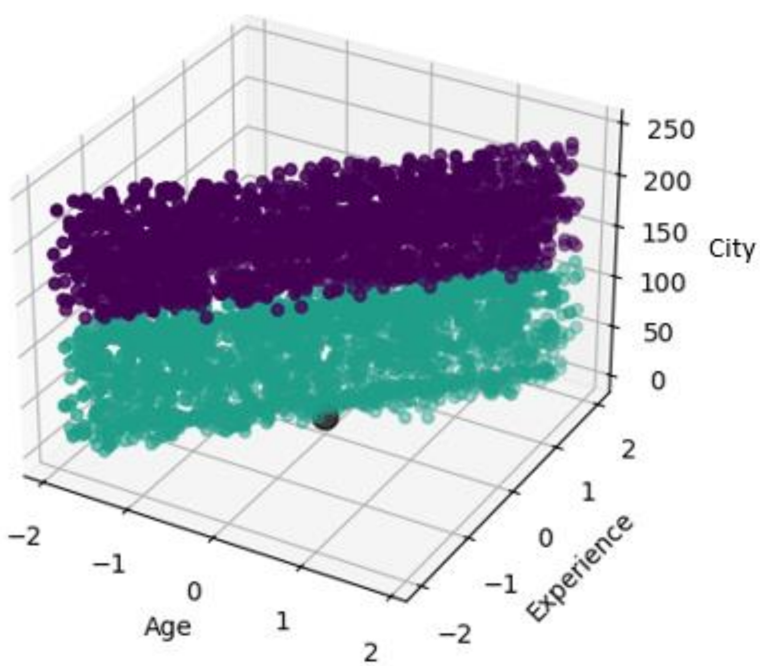
A.0 Model Performance Table

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC	Config / Notes
OLS (Baseline)	92.9%	90.3%	29.2%	44.1%	0.942	Linear baseline
Ridge	92.9%	90.3%	29.2%	45.8%	0.942	Selected regularized
Lasso	92.7%	96.0%	25.0%	44.4%	0.936	Highest precision
Elastic Net	92.9%	93.1%	28.1%	43.2%	0.941	Lowest MSE
SVM	97.6%	90.9%	83.3%	87.0%	0.989	C=10, γ=0.1
CART	97.7%	86.1%	90.6%	88.3%	0.978	depth=7, min_leaf=10
KNN	95.4%	91.7%	57.3%	70.5%	0.887	K=3, AUC gap 0.109 ✗
Random Forest	98.2%	94.3%	86.5%	90.2%	0.993	Resistant to overfitting
Gradient Boosting	98.6%	92.7%	92.7%	92.7%	0.994	Selected — best F1+AUC
CNN	95.3%	77.5%	71.9%	74.6%	0.948	Learning complex interaction
Logistic Regression	94.7%	83.1%	56.2%	67.1%	0.950	Minimize risky approvals

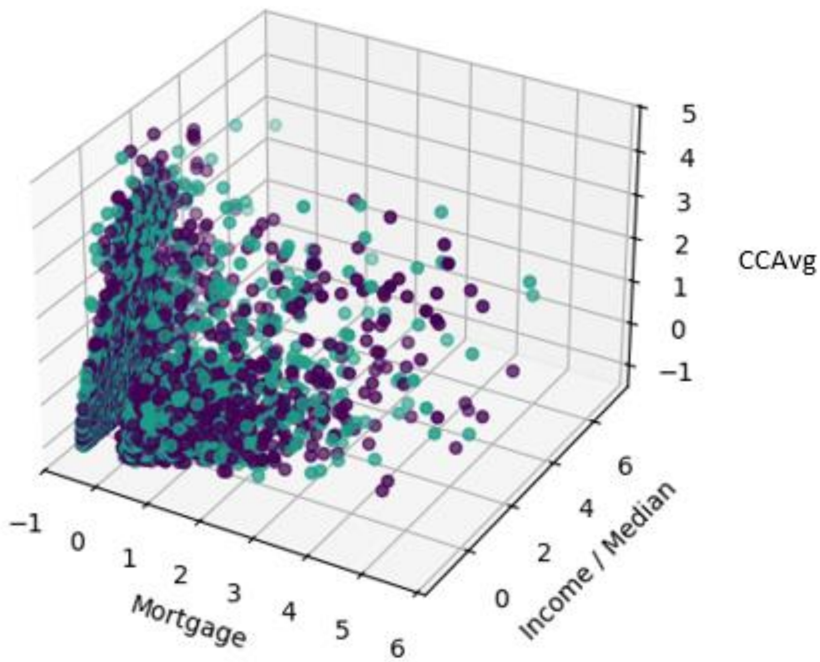
A.1 Screen Plot



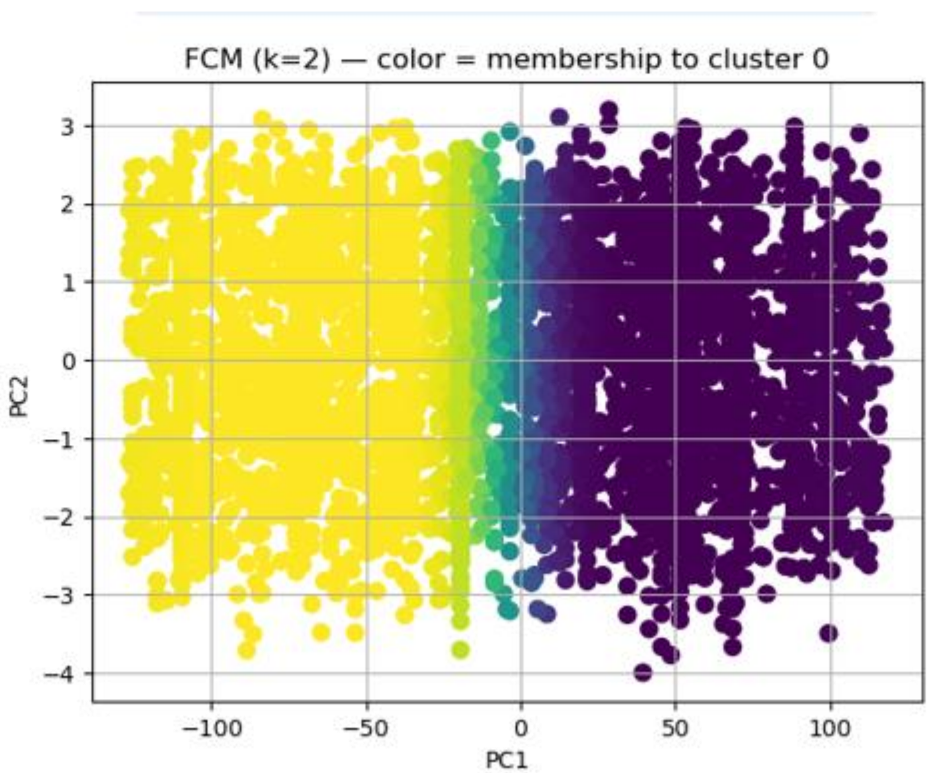
A.2 Demographic K Means Clustering



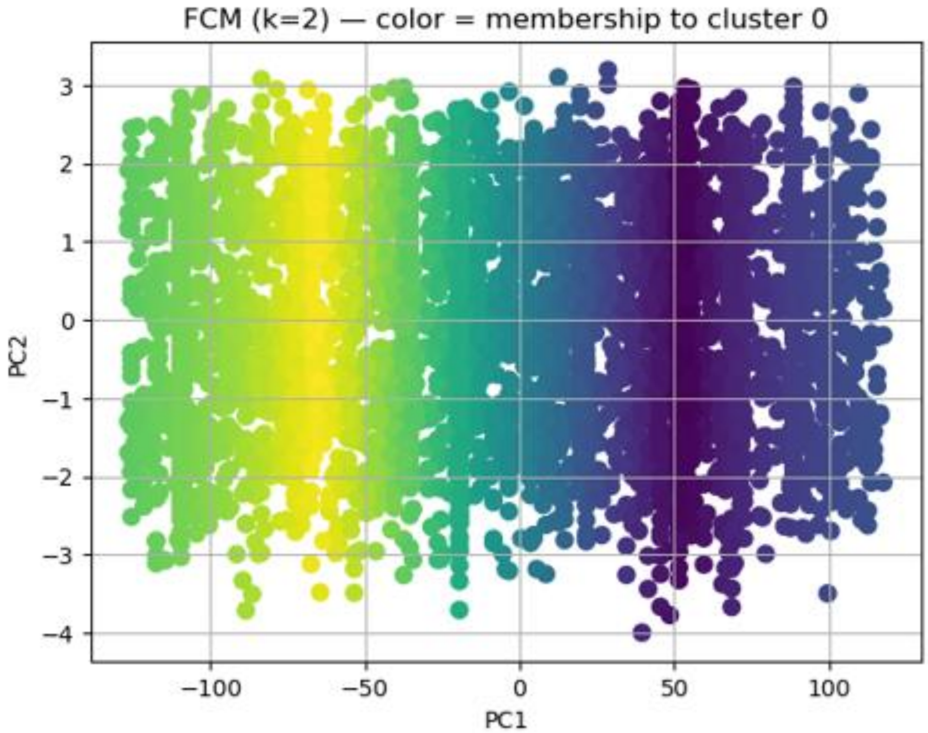
A.3 Financial K Means Clustering



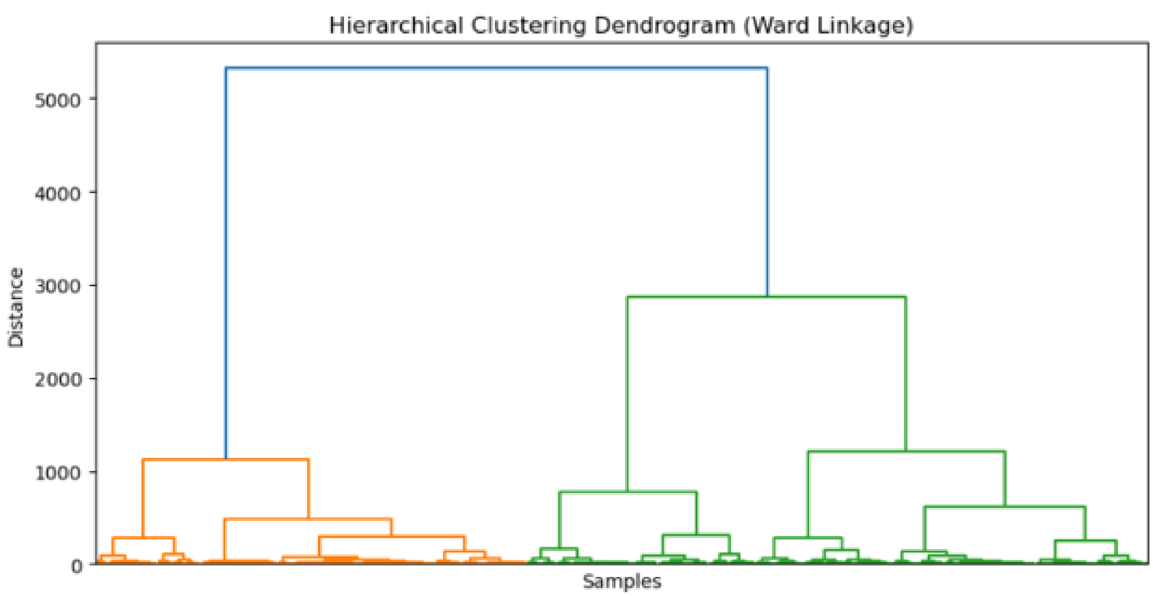
A.4 Fuzzy Clustering: m= 1.5



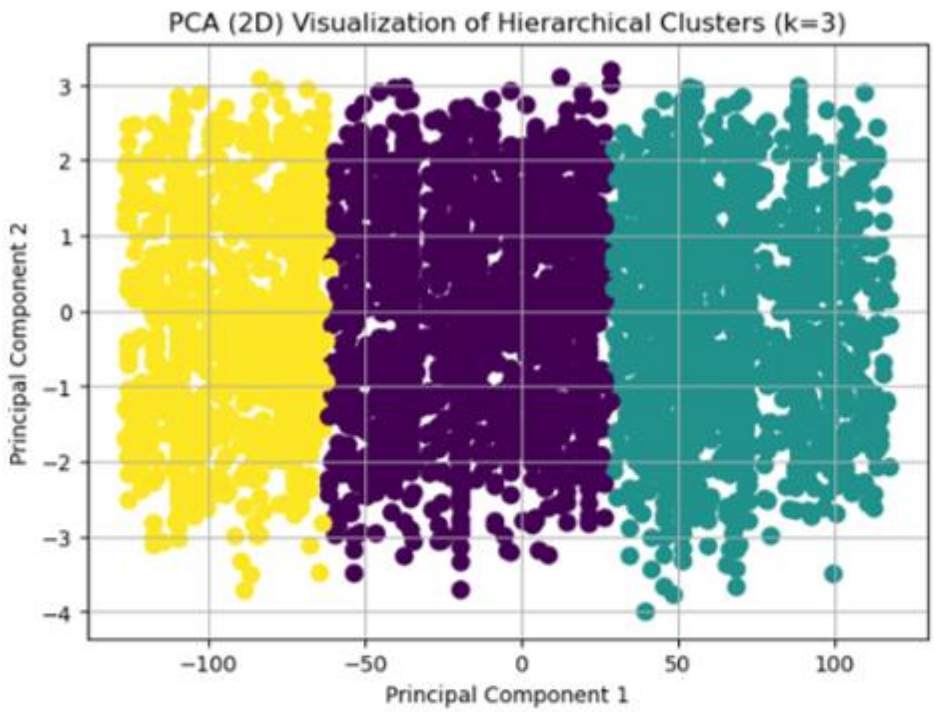
A.5 Fuzzy Clustering: m= 3.0



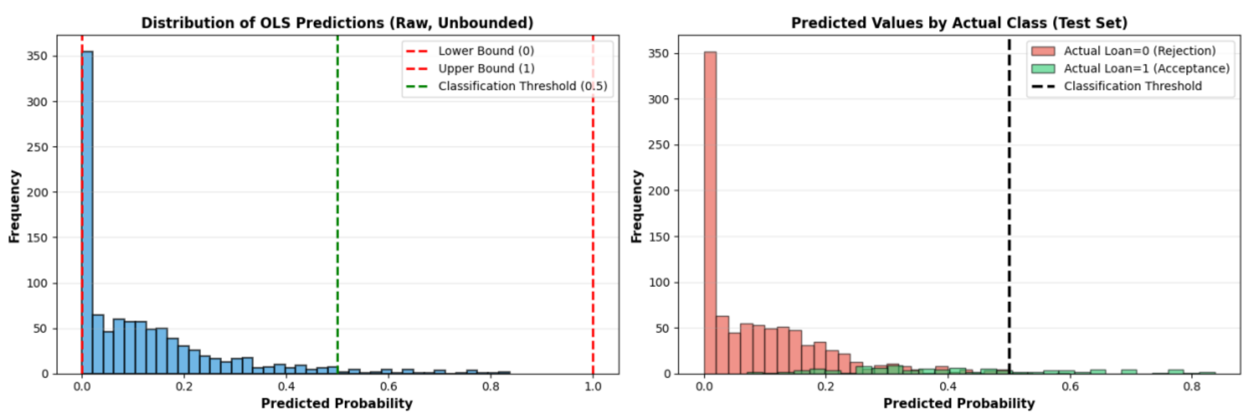
A.6 Hierarchical Clustering: Dendogram



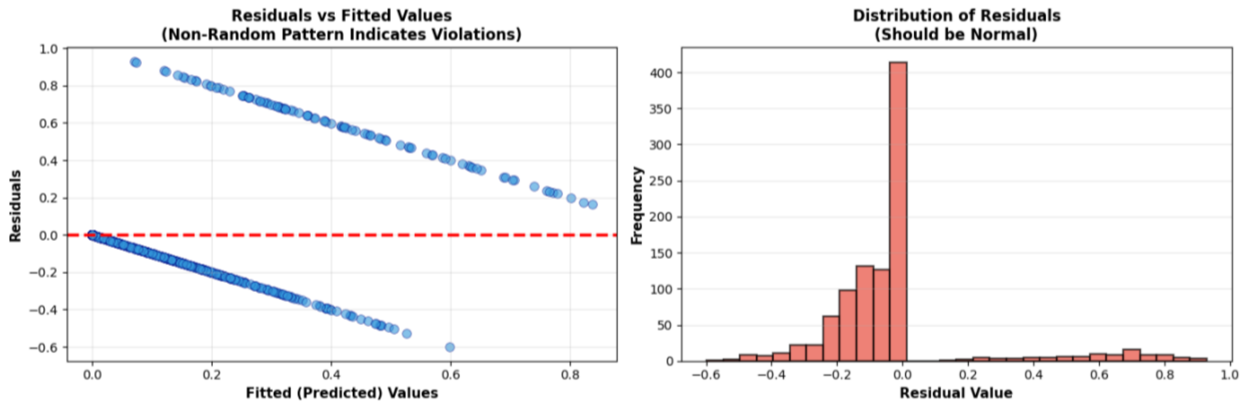
A.7 Hierarchical Clustering Plot



A.8 OLS: Prediction distribution plot



A.9 OLS: Residuals plot



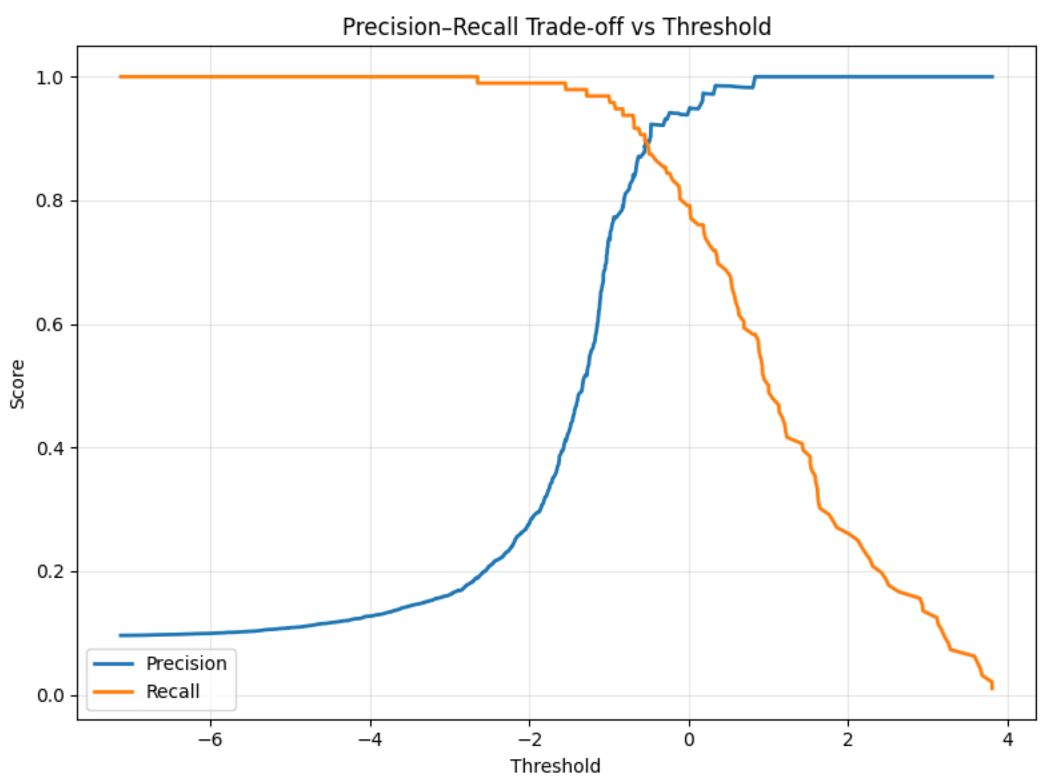
A.10 Regularized Regression Models: Features Coefficients

Feature	Ridge	Lasso	Elastic Net
Income / Median in city	0.116548	0.101928	0.112604
CD.Account	0.088649	0.070888	0.084680
Education	0.062395	0.049988	0.058279
Experience	0.052503	0	0
Age	-0.051734	0	0
Median Income Per City	0.049635	0.034404	0.045927
CCAvg	0.041459	0.039857	0.041306
Family	0.033711	0.021906	0.030887
CrediCard	-0.021830	-0.006709	-0.018517
Securities.Account	-0.021005	-0.004861	-0.017579
Online	-0.012300	0	-0.009676
Mortgage	0.011094	0.004998	0.010071

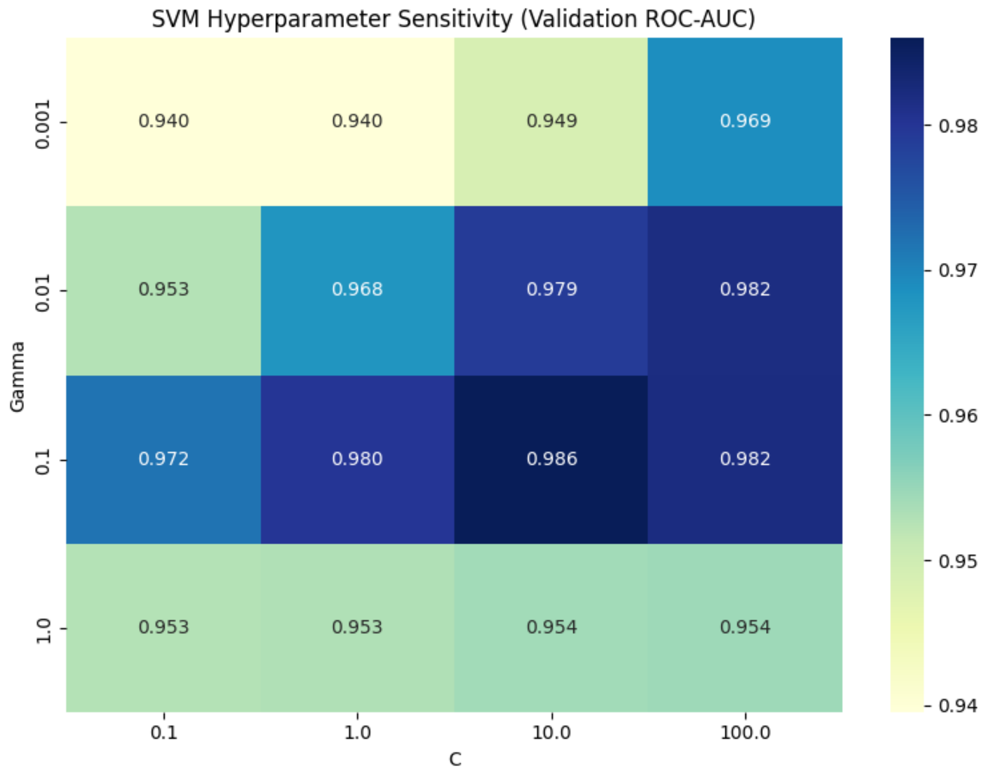
A.11 Regularized Regression Models: Hyperparameter Tuning Tables

MODEL	alpha	Train MSE	l1_ratio
Ridge	10	0.053251	-
Lasso	0.01	0.055928	-
Elastic Net	0.01	0.053793	0.2

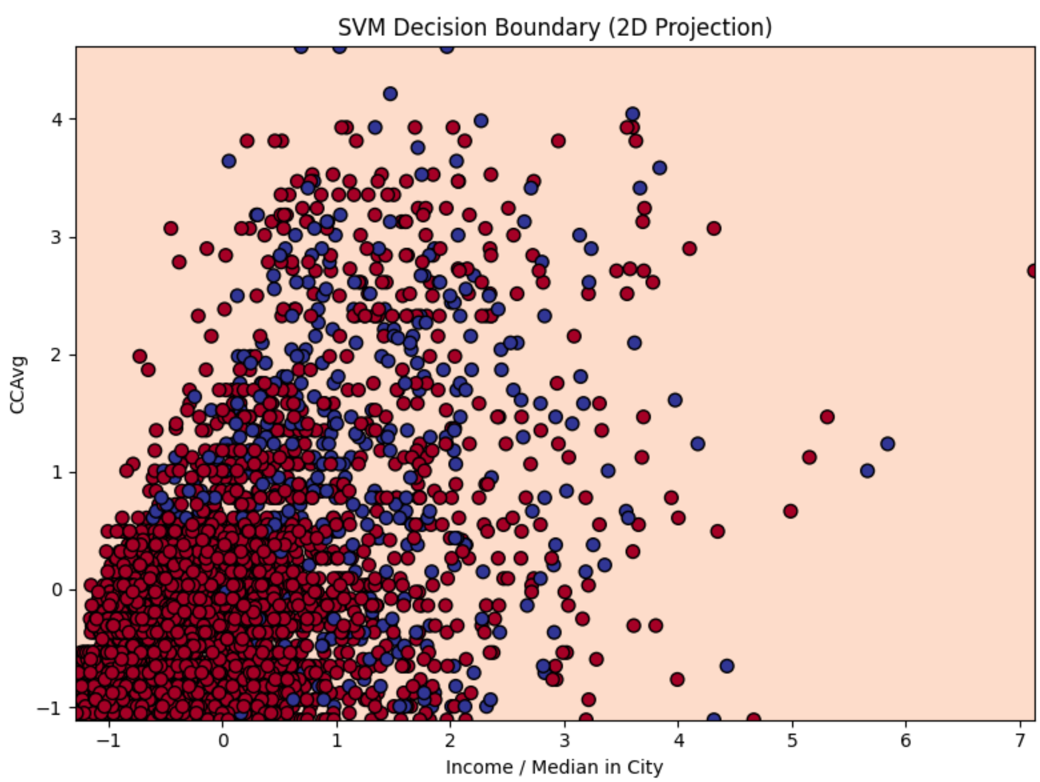
A.12 SVM: Threshold Precision-Recall Trade-off



A.13 SVM: Hyperparameter Sensitivity (ROC-AUC)

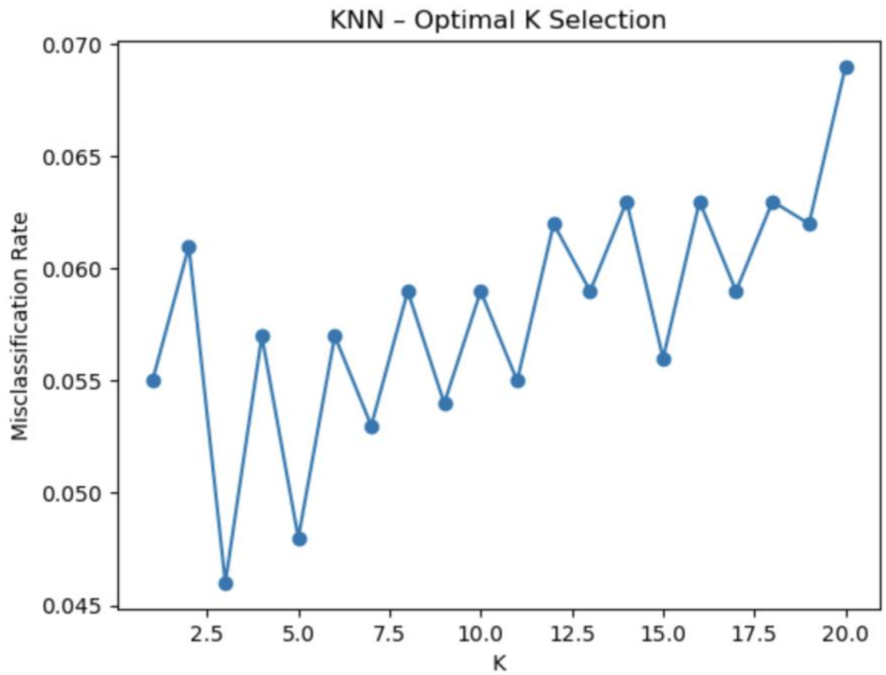


A.14 SVM: Decision Boundary (CCAvg – Income/Median in City)

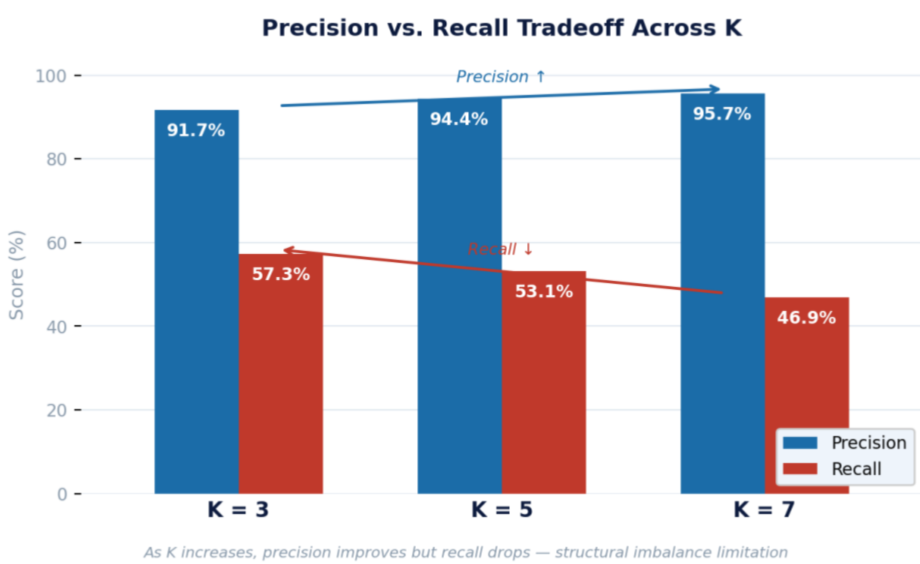


A.15 Optimal K Selection

```
Optimal K: 3
```



A.16



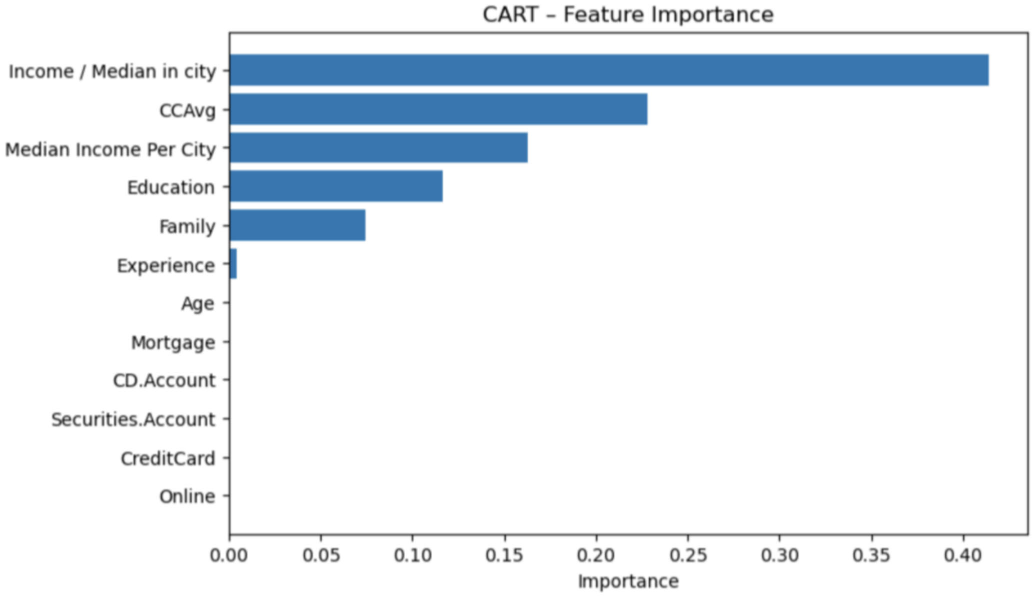
A.17 KNN Performance Table

Metric	K=3 (SELECTED)	K=5	K=7
Accuracy	95.4%	95.2%	94.7%
Precision	91.7%	94.4%	95.7%
Recall	57.3%	53.1%	46.9%
ROC AUC	0.887	0.921	0.942
AUC Gap	0.109 ✗	0.074	0.050

A.18 CART Performance Table

Metric	Baseline (depth=5)	depth=3, min_leaf=10	depth=7, min_leaf=10 ★ SELECTED
Accuracy	96.0%	94.3%	97.7%
Precision	75.5%	82.0%	86.1%
Recall	86.5%	52.1%	90.6%
ROC AUC	0.982	0.941	0.978
F1 Score	0.806	0.637	0.883

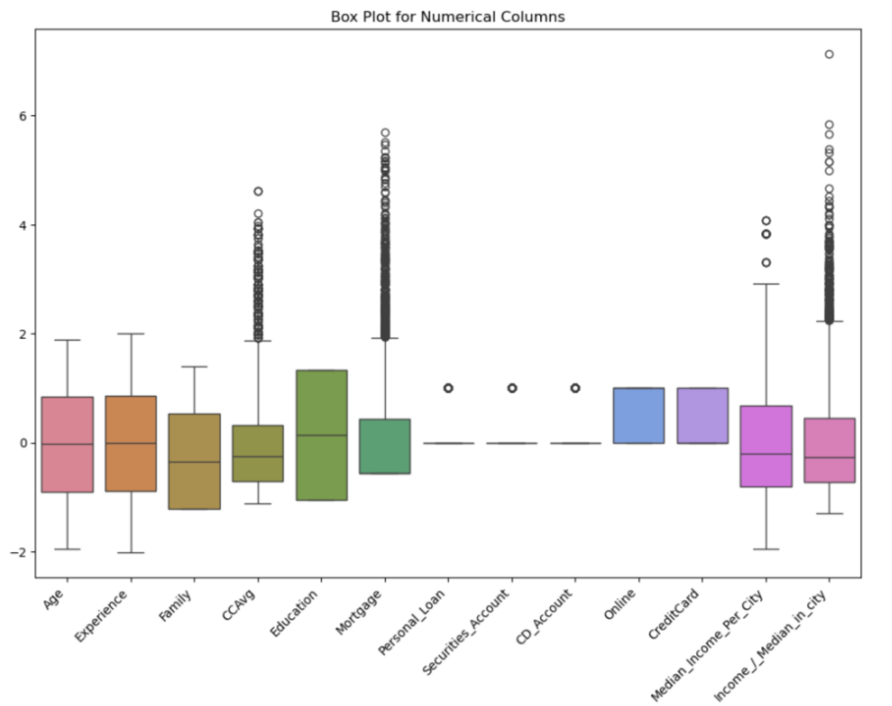
A.19 CART Feature Importance



A.20 Random Forest Accuracy

```
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Actual: 0, Predicted: 0
Random Forest Accuracy: 0.98
```

A.21 Random Forest Box Plot



A.22 CNN with 30 Epochs

```
X shape: (5000, 12)
y shape: (5000,)
Unique y values: [0 1]
Accuracy: 0.947
Precision: 0.8307692307692308
Recall: 0.5625
F1 Score: 0.6708074534161491
ROC-AUC: 0.9495529129793512

Confusion Matrix:
[[893  11]
 [ 42  54]]
```

A.23 Gradient Boosting Metrics and Confusion Matrix

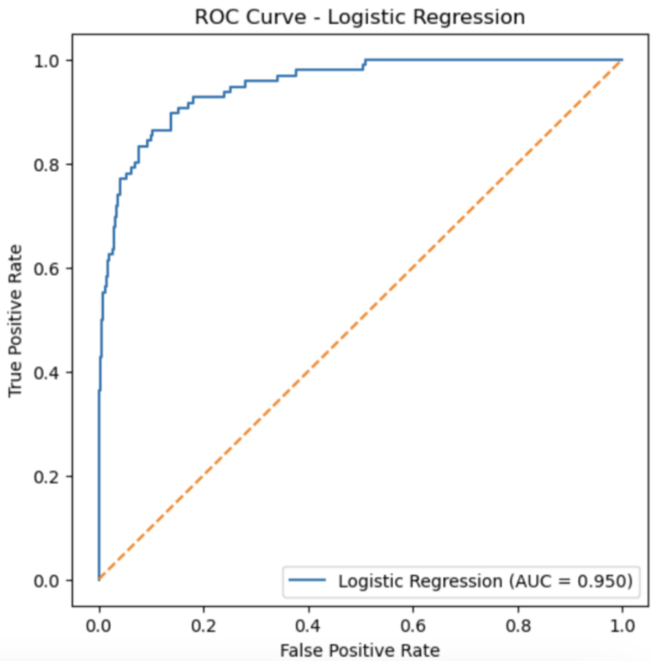
```
Accuracy: 0.986
Precision: 0.9270833333333334
Recall: 0.9270833333333334
F1 Score: 0.9270833333333334
ROC-AUC: 0.9937488477138644

Confusion Matrix:
[[897   7]
 [  7  89]]
```

A.24 Logistic Regression Metrics and Confusion Metric

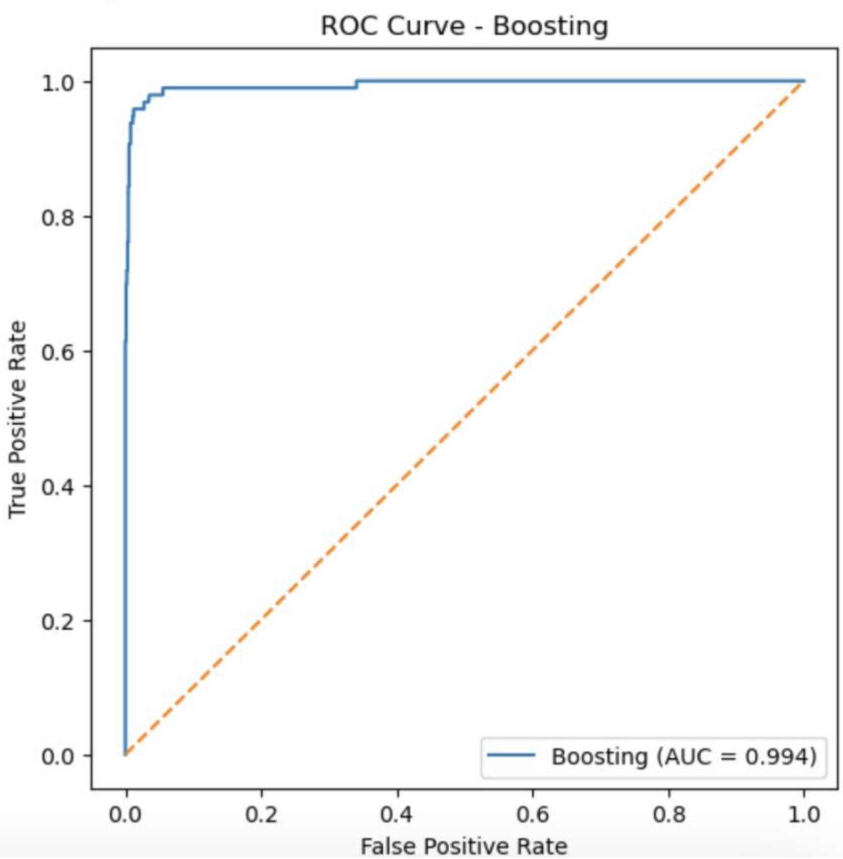
```
Accuracy: 0.937
Precision: 0.7894736842105263
Recall: 0.46875
F1 Score: 0.5882352941176471
ROC-AUC: 0.8922727691740413
Confusion Matrix:
[[892  12]
 [ 51  45]]
```

```
Logistic Regression AUC: 0.9495529129793512
```



A.25 ROC Curve – Gradient Boosting

```
Boosting AUC: 0.9937488477138644
```



A.26 Using Sigmoid in both hidden and output layers but applying different thresholds

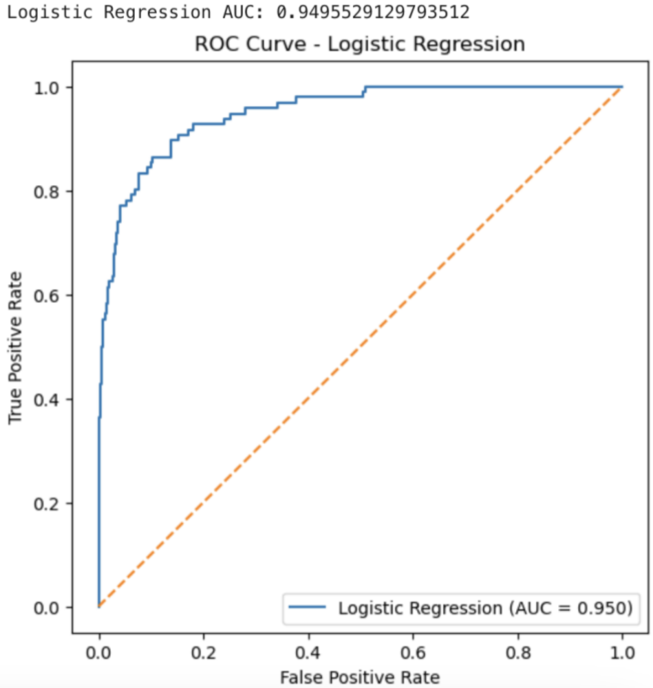
```
Threshold: 0.2
Accuracy: 0.593
Precision: 0.18200408997955012
Recall: 0.9270833333333334
F1 Score: 0.30427350427350425
ROC-AUC: 0.885520372418879
[[504 400]
 [  7  89]]

Threshold: 0.3
Accuracy: 0.741
Precision: 0.25227963525835867
Recall: 0.8645833333333334
F1 Score: 0.3905882352941176
ROC-AUC: 0.885520372418879
[[658 246]
 [ 13  83]]

Threshold: 0.4
Accuracy: 0.798
Precision: 0.29615384615384616
Recall: 0.8020833333333334
F1 Score: 0.43258426966292135
ROC-AUC: 0.885520372418879
[[721 183]
 [ 19  77]]

Threshold: 0.5
Accuracy: 0.844
Precision: 0.35436893203883496
Recall: 0.7604166666666666
F1 Score: 0.48344370860927155
ROC-AUC: 0.885520372418879
[[771 133]
 [ 23  73]]
```

```
Logistic Regression AUC: 0.9495529129793512
```



A.27 Model performance comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	94.7%	83.1%	56.2%	67.1%	0.950
Random Forest	98.2%	94.3%	86.5%	90.2%	0.993
Gradient Boosting ★	**98.6%**	**92.7%**	**92.7%**	**92.7%**	**0.994**
CNN	95.3%	77.5%	71.9%	74.6%	0.948